

1.

- a. The pseudocode for keeping track of patient info would be
 - i. name = "name"
 - ii. species = "species"
 - iii. breed = "breed"
 - iv. age = "age"
 - v. weight = "weight"
 - vi. vaccinations = "vaccinations"
 - vii. patienthistory = "patient history"
 - viii. allergies = "allergies"
 - ix. medications = "medications"
 - x. owner = "owner"
 - xi. phone = "phone"
- b. The variables that would be required are the name, species, weight, owner, and the phone number as these are important for the patient. For the optional; breed, age, vaccinations, patient history, allergies, and medications would be the optional information. And to categorize the information; name, species, breed, and owner would be non-mutable while age, weight, vaccinations, patient history, allergies, medications, and the phone number would be the mutable as these can change at any time.
- c. For patients that check in for an appointment, the best solution would be to create a "first in, first out" type of data structure so the code would need to have date, time imported to keep track of the time. The next thing to look for is the status of the patient to see if they're waiting, seeing a doctor, or if they've been seen and leaving.
- d. For the emergency visits, the only thing that would be added to the considerations in part c would be to add an immediate or priority care structure so that whatever emergency patient can be seen first.

2.

- a. The considerations for the training AI would be to correctly label all the images and the test samples so that the AI will not misidentify some of the images and use them in the network evaluation, the other thing that needs to be considered would be the identifying each animal on each of the images so that an image of a cat will not be mislabeled as a dog. The bottleneck for this instance would be the processing load and time as the sample size can be done through many epochs.
- b. Like the response before, the images and the test samples would need to be properly labeled and identified so making a list for test images and training images would be one of the things to be considered. The next thing would be to label all the images in the training list to correctly identify each animal.
- c. Again, taking the response from before, keeping the lists separate would make the most sense as you can shuffle the lists separately so that a training image will not end up in the test sample or vice versa.